

# Global Optimisation of Polynomial Portfolio Risk: What the Lasserre Hierarchy Certifies

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## Abstract

I ask whether a certified global minimiser of a degree-4 portfolio risk polynomial is worth having. I solve the order-2 Lasserre relaxation, dense and with correlative sparsity, and certify a point a posteriori when its value matches the bound. On 1760 synthetic instances, one local start misses the global minimum in 45% of the cases, and the relaxation certifies 98.4% of them. On 37 US stocks (2015–2025), the certified minimiser lowers volatility and CVaR relative to equal weights, as any risk-minimising rule does. It is not better than Ledoit–Wolf minimum variance or minimum CVaR on those measures, and no net-return difference is distinguishable from zero. A Frobenius-ball formulation meant to protect against kurtosis estimation error does not beat the nominal problem, except marginally at the largest noise level; shrinkage towards a structured target does better. A first version of this work contained errors, which I corrected before running the experiments. The conclusions hold for one family of synthetic polynomials and one fixed universe.

## 1 Introduction

Portfolio risk that accounts for skewness and kurtosis is a polynomial in the weights. With the first four moments the objective has degree four and is not convex in general. A local solver returns a stationary point. The Lasserre hierarchy [7] returns a lower bound, and a feasible point whose value is close to that bound is globally optimal. I ask what this certificate is worth on portfolio risk polynomials.

I do three things. I implement the order-2 relaxation, dense and with correlative sparsity [13], and a certificate that does not depend on a rank test. I compare it with multi-start local search on 1760 synthetic instances (Experiment A). I test a Frobenius-ball formulation for estimation error on the kurtosis coefficients (Experiment B), and the certified minimiser on 37 US stocks (Experiment C).

Three results stand out. Local search fails more often than I expected, even on the default synthetic regime. With one start it misses the global minimum in 45% of the instances, and in 78% of those with  $n \geq 30$  assets. The certificate is available in most cases, but its cost at  $n = 50$  exceeds that of 100 local runs. On real data the certified minimiser differs from the local one on few dates. It brings no out-of-sample gain over the risk-based baselines.

The first version of this work (tag `v1.0-paper`) contained errors. The sparse budget constraint decoupled the cliques, a size-reduction ratio mixed two metrics, a dense timing was unsupported, and a benchmark described as non-convex is convex. Appendix A lists them with the measured values. I fixed them before running any experiment below.

## 2 Risk model

The synthetic objective is

$$f(x) = x^\top \Sigma x + \sum_i \eta_i x_i^3 + \sum_{i,j} \kappa_{ij} x_i^2 x_j^2 + \text{impact terms}, \quad (1)$$

on  $K = \{x \geq 0, b_{\text{lo}} \leq \mathbf{1}^\top x \leq b_{\text{hi}}\}$  with  $b_{\text{lo}} = 0.95$  and  $b_{\text{hi}} = 1$ . The coefficients  $\kappa_{ij}$  are model parameters. They are not the fourth co-moment tensor of a portfolio, which would be a dense contraction  $\sum \kappa_{ijkl} x_i x_j x_k x_l$ . Cross terms exist only within clusters, so the interaction graph is block-structured. The markets are synthetic and uncalibrated in every regime of Experiment A.

On market data (Experiment C) the objective is estimated on a window of 756 days:  $F(x) = \ell_2 m_2(x) - \ell_3 m_3(x) + \ell_4 m_4(x)$ , with  $m_k(x)$  the sample moment of order  $k$  of the centred portfolio return  $x^\top \tilde{r}_t$ . The weights  $\ell_k$  are frozen in advance so that each term contributes the same amount on a single-asset portfolio. There is no return term: the comparison is about risk. A sector variant keeps only intra-sector co-moments, which is a modelling choice and not a property of the data.

### 3 Relaxations and certificates

The relaxation of order  $d$  replaces each monomial  $x^\alpha$  with  $|\alpha| \leq 2d$  by a variable  $y_\alpha$ . It asks that the moment matrix  $M_d(y)$  and the localising matrices of the constraints be positive semidefinite, and minimises  $\sum_\alpha f_\alpha y_\alpha$ . The optimum  $\lambda_{\text{lb}}$  satisfies  $\lambda_{\text{lb}} \leq f^*$ . Since  $f$  has degree four,  $d = 2$  is the smallest usable order. Under an Archimedean condition  $\lambda_{\text{lb}}$  converges to  $f^*$  as  $d$  grows [7, 11]. Convergence at  $d = 2$  is not guaranteed.

**Correlative sparsity.** Let  $I_1, \dots, I_m$  be the maximal cliques of a chordal extension of the interaction graph. The sparse relaxation uses one moment matrix per clique [13, 8]. Every sparse constraint is implied by the dense ones at the same order, so I expect  $\lambda_{\text{lb}}^{\text{sparse}} \leq \lambda_{\text{lb}}^{\text{dense}}$ , and I test it.

The budget  $\mathbf{1}^\top x \in [b_{\text{lo}}, b_{\text{hi}}]$  involves every variable and belongs to no clique. Version 1 replaced it by a constraint on  $E[\sum x]$ , which decoupled the cliques. I now add the sector sums  $s_k = \sum_{i \in O_k} x_i$ , where  $O_k$  is the set of variables owned by clique  $k$ . The equalities  $s_k = \sum_{i \in O_k} x_i$  live in the clique  $I_k \cup \{s_k\}$  and the budget lives in a clique on  $\{s_1, \dots, s_m\}$ . This is the star variant. For many cliques I use partial sums  $p_k = p_{k-1} + s_k$ , which give cliques of size three (chain variant). Both satisfy the running intersection property.

**Certificate.** After solving, I take candidate points from the Henrion–Lasserre extraction [6] when the rank condition holds, and from the first-order moments otherwise. I polish them with a local solve and check feasibility to  $10^{-8}$ . A point  $\hat{x}$  is certified when  $f(\hat{x}) - \lambda_{\text{lb}} \leq \max(10^{-6}, 10^{-4}|\lambda_{\text{lb}}|)$ . Since  $\lambda_{\text{lb}} \leq f^* \leq f(\hat{x})$ , this holds whatever the rank of the moment matrix. The flat-extension test of Curto and Fialkow [1] is reported as a secondary diagnostic. Solves use CLARABEL [5] through CVXPY [4] with tolerance  $10^{-7}$ . The bound is only as accurate as the solver; Appendix B measures the accuracy.

### 4 Expectations and measurements

Before the experiments I wrote down what I expected, from theory or from the claims of version 1. Table 1 compares these expectations with the measurements. Each line is detailed in the section indicated.

## 5 Experiments

### 5.1 A two-asset check

The 2-asset benchmark of version 1 has a positive definite Hessian on  $K$  (smallest eigenvalue 1.68) and a single local minimum. It validates the code but illustrates nothing about local minima. I built a non-convex instance by raising the skewness scale until a multi-start run finds two minima

Table 1: Expectations against measurements. “Held” means the measurement agrees with the expectation on the tested instances.

| Expectation                                  | Reason                  | Measurement                                                                                                                                                                                      | Verdict                   | Explanation                                                                                     |
|----------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------|
| Sparse bound equals dense bound              | Theorem 7 of version 1  | Version 1: $-22.1$ against $-2.43$ on one instance. After the budget fix, median dense minus sparse $2.9 \times 10^{-10}$ , but 5 of 1100 instances exceed $10^{-3}$ (max $9.0 \times 10^{-3}$ ) | Partly                    | Budget bug in version 1; remaining gaps are on instances the sparse relaxation does not certify |
| 2-asset benchmark is non-convex              | Version 1 caption       | Smallest Hessian eigenvalue on $K$ : $+1.68$ ; one minimum                                                                                                                                       | Not held                  | The benchmark is convex; a non-convex instance has to be built (Sec. 5.1)                       |
| Local search is enough in the default regime | Protocol of Exp. A      | Several local minima in 34% of default-regime instances; one start fails with probability 0.30 at $n = 15$ and 0.83 at $n = 50$                                                                  | Not held                  | The number of local minima grows with $n$ ; I did not test why                                  |
| Order-2 certificate in most cases            | Tightness of low orders | 98.4% certified in total; 74% for the default regime at $n = 50$                                                                                                                                 | Held, except at large $n$ | 79 solves return an inaccurate status                                                           |
| Flat extension at $d = 2$ certifies          | Version 1               | Flat in 97.4% of instances; at $d = 3$ on the non-convex pair the ranks are 5 and 1 with the same bound                                                                                          | Unstable                  | The solver may return a point inside the optimal face; not tested                               |
| Size reduction of 398 at $n = 50$            | Version 1 table         | 128 and 215 (PSD entries), 104 and 135 (moments), for star and chain                                                                                                                             | Corrected                 | Version 1 compared two different metrics                                                        |
| Dense SDP unaffordable at $n = 50$           | “> 60 h” in version 1   | Measured: 193 s and 4.7 GB at $n = 15$ . Extrapolated, not measured: $3.8 \times 10^6$ s at $n = 50$                                                                                             | Not tested                | Extrapolation from five points                                                                  |
| SDP faster than local search                 | Cost of one solve       | At $n = 50$ in the default regime, 80 s against 33 s for 100 local runs                                                                                                                          | Not held                  | The sparse SDP still has blocks of size up to 66                                                |
| Ball formulation helps under kurtosis error  | Experiment B design     | Worse than nominal for $\rho \leq 0.2$ ; 5% better at $\rho = 0.5$ with $\gamma = 0.1$                                                                                                           | Not held                  | The ball adds a bias proportional to $\delta$ ; not otherwise tested                            |
| Certified minimiser better out of sample     | Motivation              | Net-return intervals contain zero in all 13 comparisons; volatility and CVaR are above those of minimum variance and minimum CVaR                                                                | Not held                  | Moment estimation error, not separated from model error; Sec. 5.5                               |

(smallest Hessian eigenvalue  $-3.46$ ). The best local minimum has value  $-0.9849$  and the other  $-0.5640$ ; 41% of the local starts end in the worse one. The order-2 bound is  $-0.9849$  and the certificate gap is  $3.2 \times 10^{-9}$ . At  $d = 3$  the bound is the same to  $10^{-8}$ , but the rank test fails (ranks 5 and 1). So the certificate is available at  $d = 2$  even though the rank test, applied at a higher order, would reject it.

## 5.2 Experiment A: local search against the relaxation

I generate 1760 instances with  $n \in \{2, 4, 6, 8, 10, 15, 30, 50\}$  assets in three regimes (default, mixed, stress), 50 to 120 per cell. For each instance I solve the sparse relaxation at  $d = 2$  (the dense one for  $n \leq 10$ ) and run 100 SLSQP starts drawn i.i.d. on the budget set. The reference optimum  $f_{\text{cert}}$  is the certified point when there is one, and the best of 100 runs otherwise. A search with  $K$  starts fails when its best value exceeds  $f_{\text{cert}}$  by more than  $\max(10^{-6}, 10^{-4}|f_{\text{cert}}|)$ .

*Expected.* In the default regime I expected local search to find the global minimum, and in the stress regime I expected failures.

*Observed.* Several local minima appear in 34%, 75% and 81% of the default, mixed and stress instances.

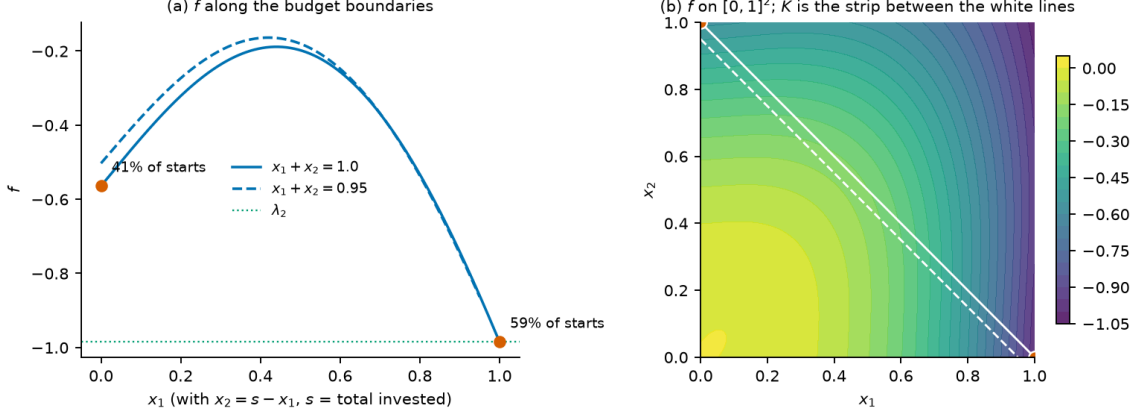


Figure 1: Non-convex 2-asset instance. (a)  $f$  along the two budget boundaries; dots are the local minima with the share of starts that reach them. (b)  $f$  on the unit square;  $K$  is the strip between the white lines.

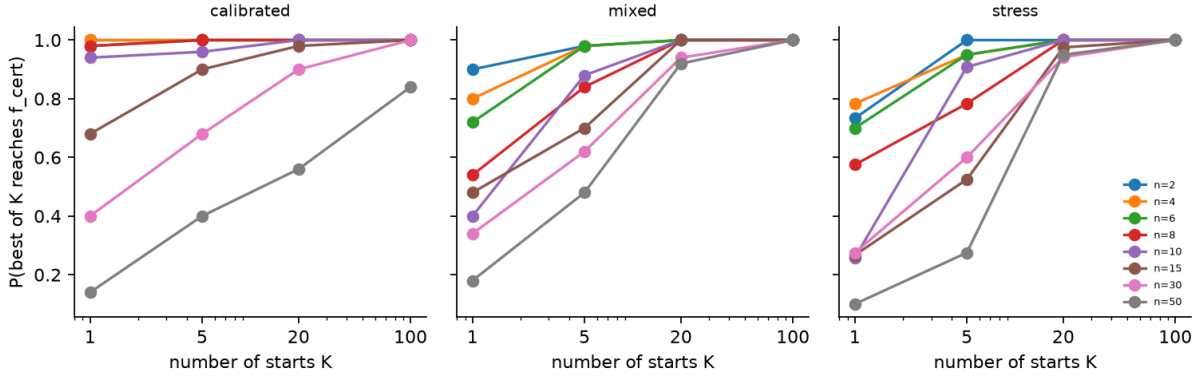


Figure 2: Experiment A. Share of instances where the best of  $K$  local starts reaches the reference optimum, by regime and number of assets  $n$ .

The probability that one start fails grows with  $n$  in all regimes (Table 2). In the default regime it is 0.30 at  $n = 15$  and 0.83 at  $n = 50$ .

More starts help without closing the gap (Figure 2). Over all instances the best of one start fails in 45% of the cases and the best of 100 in 0.5%. In the default regime at  $n = 50$ , 100 starts still fail in 16% of the instances.

*Reading.* The default regime is not an easy regime beyond about ten assets. I did not test why local minima multiply with  $n$ .

The relaxation certifies 98.4% of the 1760 instances, but unevenly: 74% in the default regime at  $n = 50$ , and 29 instances in total are not certified. 79 sparse solves return an inaccurate status. I also solved the dense relaxation on 1100 instances. The difference  $\lambda_{\text{lb}}^{\text{dense}} - \lambda_{\text{lb}}^{\text{sparse}}$  has median  $2.9 \times 10^{-10}$ . Only 5 instances exceed  $10^{-3}$  (maximum  $9.0 \times 10^{-3}$ ). All five are non-certified instances with  $n = 4$  or  $n = 8$ : there the sparse relaxation is weaker. The smallest difference is  $-1.0 \times 10^{-6}$ , which I take to be solver accuracy. Across all instances, the sparse bound exceeds the best local value by at most  $9.4 \times 10^{-6}$ , again the order of the solver tolerance.

On time, the sparse relaxation is not cheaper than local search at the largest size. At  $n = 50$  the median solve takes 80 s in the default regime against 33 s for 100 local runs, and 20 s against 2 s in the stress regime. These are wall times from six concurrent workers and are indicative only.

Table 2: Experiment A, selected cells. Multi-min: share of instances with several distinct local minima among 100 runs;  $p_1$ : mean probability that one start misses the reference optimum; fail  $K$ : share of instances where the best of  $K$  starts misses it; cert.: share certified a posteriori at  $d = 2$ ; times are medians of the sparse SDP and of 100 local runs, in seconds.

| regime     | $n$ | inst. | multi-min | $p_1$ | fail 20 | fail 100 | cert. | SDP (s) | 100 local (s) |
|------------|-----|-------|-----------|-------|---------|----------|-------|---------|---------------|
| calibrated | 4   | 50    | 0%        | 0.00  | 0%      | 0%       | 100%  | 0.1     | 0.7           |
| calibrated | 10  | 50    | 18%       | 0.09  | 0%      | 0%       | 100%  | 0.6     | 1.4           |
| calibrated | 15  | 50    | 48%       | 0.30  | 2%      | 0%       | 100%  | 0.7     | 1.9           |
| calibrated | 30  | 50    | 92%       | 0.65  | 10%     | 0%       | 100%  | 2.3     | 3.9           |
| calibrated | 50  | 50    | 98%       | 0.83  | 44%     | 16%      | 74%   | 79.9    | 32.7          |
| mixed      | 4   | 50    | 52%       | 0.23  | 0%      | 0%       | 86%   | 0.1     | 0.5           |
| mixed      | 10  | 50    | 84%       | 0.52  | 0%      | 0%       | 100%  | 0.5     | 0.8           |
| mixed      | 15  | 50    | 92%       | 0.66  | 0%      | 0%       | 94%   | 0.6     | 0.9           |
| mixed      | 30  | 50    | 100%      | 0.77  | 6%      | 0%       | 96%   | 1.9     | 1.7           |
| mixed      | 50  | 50    | 100%      | 0.86  | 8%      | 0%       | 98%   | 67.4    | 11.9          |
| stress     | 4   | 120   | 65%       | 0.29  | 0%      | 0%       | 99%   | 0.1     | 0.4           |
| stress     | 10  | 120   | 80%       | 0.59  | 0%      | 0%       | 100%  | 0.5     | 0.6           |
| stress     | 15  | 120   | 85%       | 0.69  | 2%      | 0%       | 100%  | 0.6     | 0.7           |
| stress     | 30  | 120   | 100%      | 0.85  | 6%      | 0%       | 100%  | 1.8     | 1.1           |
| stress     | 50  | 120   | 100%      | 0.90  | 5%      | 0%       | 100%  | 19.7    | 2.1           |

Table 3: Dense versus sparse relaxation, order  $d = 2$ , one synthetic instance per  $n$ . Dense times and memory are measured up to  $n = 15$ ; rows marked \* are log-log extrapolations from five measured points, not measurements (95% prediction intervals in `results/dense_extrapolation.csv`). Sparse figures use the chain budget.

| $n$ | dense block | dense (s)           | dense (GB) | dense status       | sparse block | sparse (s) |
|-----|-------------|---------------------|------------|--------------------|--------------|------------|
| 4   | 15          | 0.07                | 0.12       | optimal            | 6            | 0.07       |
| 6   | 28          | 0.21                | 0.13       | optimal            | 10           | 0.09       |
| 8   | 45          | 1.62                | 0.19       | optimal            | 15           | 0.13       |
| 10  | 66          | 6.12                | 0.39       | optimal            | 21           | 0.50       |
| 12  | 91          | 40.62               | 1.06       | optimal inaccurate | 15           | 0.26       |
| 15  | 136         | 192.81              | 4.67       | optimal inaccurate | 21           | 0.71       |
| 30  | 496         | $5.3 \times 10^4$ * | 58*        | extrapolated       | 21           | 6.6        |
| 50  | 1326        | $3.8 \times 10^6$ * | 546*       | extrapolated       | 21           | 26.3       |

### 5.3 Size and cost of the relaxation

Table 3 gives measured and extrapolated costs. Up to  $n = 15$  the dense relaxation is measured: 193 s and 4.7 GB, against 0.71 s for the sparse one on a similar instance. At  $n = 12$  and  $n = 15$  the dense solve ends with an inaccurate status.

Beyond  $n = 15$  the dense figures are extrapolations. The fit has an exponent of 4.35 on the block size and five fitting points. Its interval at  $n = 50$  spans  $6.2 \times 10^5$  to  $2.4 \times 10^7$  s, and I report it as such.

Take  $n = 50$  with ten cliques of five assets. The total number of PSD entries falls by a factor of 128 (star budget) to 215 (chain). The number of free moments falls by 104 to 135. The sparse solves take 26 to 51 s.

Version 1 reported 398 by counting only moment blocks. It left out the localising matrices and the budget clique, which it did not impose. The same moment-block metric on the corrected relaxation gives 201 (star) and 371 (chain). Version 1 also reported 1.85 s for the solve.

Table 4: Experiment B, mean true regret over 200 draws, mixed markets. Nominal:  $\delta = 0$ ; ball:  $\delta = \gamma \|\theta\|_W$  with  $\gamma = 0.1$  and 1; rule:  $\delta = c \hat{\sigma}$  with  $c = 1$ ; shrinkage towards the structured target with  $s = 0.5$  and 1.

| $n$ | $\rho$ | nominal              | ball 0.1             | ball 1               | rule                 | shrink 0.5           | shrink 1             |
|-----|--------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| 10  | 0.05   | $1.8 \times 10^{-7}$ | $1.3 \times 10^{-6}$ | $5.5 \times 10^{-5}$ | $4.6 \times 10^{-7}$ | $1.1 \times 10^{-6}$ | $4.5 \times 10^{-6}$ |
| 10  | 0.2    | $2.8 \times 10^{-6}$ | $3.5 \times 10^{-6}$ | $5.5 \times 10^{-5}$ | $6.1 \times 10^{-6}$ | $1.8 \times 10^{-6}$ | $4.5 \times 10^{-6}$ |
| 10  | 0.5    | $1.8 \times 10^{-5}$ | $1.7 \times 10^{-5}$ | $6.2 \times 10^{-5}$ | $3.0 \times 10^{-5}$ | $5.7 \times 10^{-6}$ | $4.8 \times 10^{-6}$ |
| 6   | 0.05   | 0                    | $1.0 \times 10^{-3}$ | $5.9 \times 10^{-1}$ | 0                    | 0                    | 0                    |
| 6   | 0.2    | $3.5 \times 10^{-4}$ | $3.3 \times 10^{-2}$ | $5.9 \times 10^{-1}$ | $1.5 \times 10^{-1}$ | $9.6 \times 10^{-5}$ | $4.0 \times 10^{-5}$ |
| 6   | 0.5    | $7.4 \times 10^{-2}$ | $1.2 \times 10^{-1}$ | $5.9 \times 10^{-1}$ | $4.4 \times 10^{-1}$ | $3.0 \times 10^{-2}$ | $1.8 \times 10^{-2}$ |

#### 5.4 Experiment B: estimation error on the kurtosis coefficients

The ball formulation minimises  $f(x; \hat{\kappa}) + \delta \|Wz(x)\|_2$ , where  $z(x)$  stacks the monomials  $x_i^4$  and  $x_i^2 x_j^2$  and  $W$  weights the off-diagonal entries by  $\sqrt{2}$ . It is the worst case over a Frobenius ball of radius  $\delta$  around  $\hat{\kappa}$ . It is parametric robustness; it is not the moment-uncertainty set of Delage and Ye [2]. I draw four markets ( $n = 6, 10$ ; mixed and stress regimes). I perturb the true coefficient vector with an error of relative size  $\rho \in \{0.05, 0.1, 0.2, 0.5\}$ , with 200 draws per market and  $\rho$ . I compare four methods. The nominal problem has  $\delta = 0$ . The ball uses  $\delta = \gamma \|\theta\|_W$  for six values of  $\gamma$ . The automatic rule sets  $\delta = c \hat{\sigma}$  from a bootstrap of  $T = 20$  synthetic observations. Shrinkage moves  $\hat{\kappa}$  towards a structured target. The metric is the true regret  $f(\hat{x}; \kappa) - f(x^*; \kappa)$ .

*Expected.* I expected the ball to lower the regret when  $\rho$  is large and to cost little when  $\rho$  is small.

*Observed.* On the 10-asset mixed market the ball is worse than nominal for  $\rho \leq 0.2$ , and the loss grows with  $\delta$  (Figure 3, Table 4). At  $\rho = 0.05$  the mean regret is  $1.8 \times 10^{-7}$  for nominal and  $1.3 \times 10^{-6}$  for  $\gamma = 0.1$ .

At  $\rho = 0.5$  the ball with  $\gamma \leq 0.1$  lowers the mean regret by 2 to 5%, and the bootstrap intervals exclude zero. At  $\gamma = 0.2$  the difference is not distinguishable from zero, and with  $\gamma \geq 0.5$  the ball is worse again. Shrinkage with  $s = 0.75$  brings the regret to 0.21 of the nominal value. The automatic rule never beats nominal.

In the stress markets the nominal solution is a corner that coincides with  $x^*$  in at least 95% of the draws. One cell discriminates: 10 assets at  $\rho = 0.5$ , where nominal regret is  $2.2 \times 10^{-2}$ . There the ball with  $\gamma = 1$  raises it to 0.72, and shrinkage with  $s = 0.5$  brings it to 0.000.

An order-2 SDP check on 1120 solves certifies the nominal local result in 100% of the cases. It certifies the ball results in 99% of the cases for  $\gamma = 0.02$  and 25% for  $\gamma = 1$ .

*Reading.* The ball adds a penalty proportional to  $\delta$ , and here the nominal regret is already small. Shrinkage helps more, probably because the synthetic coefficients are drawn i.i.d. from narrow ranges, which makes the mean-based target close to the truth. I did not test this explanation, and the markets are synthetic.

#### 5.5 Experiment C: real data

The data are Yahoo Finance adjusted closes for 37 large US stocks in 9 sectors, 2009–2025, fixed ex post. This introduces survivorship bias. I rebalance monthly from 2015 to 2025 (132 dates) with a 756-day window, long-only weights capped at 0.25, and costs of 5 and 10 bp. Universe U10 has ten assets and the dense model. Universe U37 has all 37 assets and the sector model. The protocol was frozen before any run, with three amendments made before the evaluation period (Appendix D). Comparators are equal weights, Ledoit–Wolf minimum variance [9], equal risk contribution, hierarchical risk parity [10] and minimum CVaR [12].

*Expected.* Third and fourth sample co-moments are noisy on 756 days. I expected the global

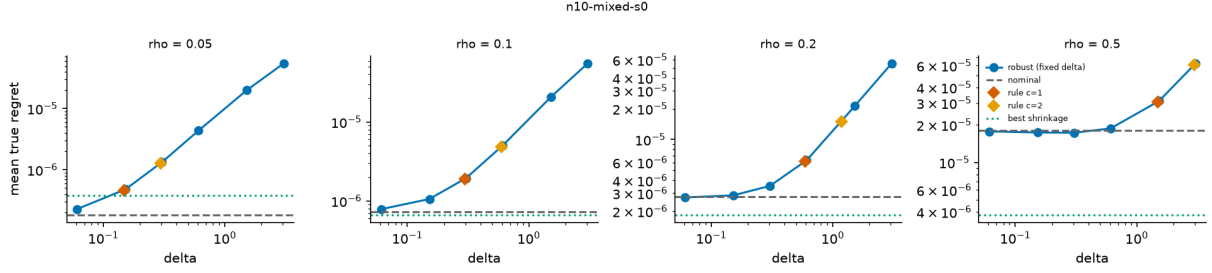


Figure 3: Experiment B, 10-asset mixed market. Mean true regret against  $\delta$  for the ball (circles), the automatic rule (diamonds), the nominal problem (dashed) and the best shrinkage (dotted), for four noise levels  $\rho$ . Log axes.

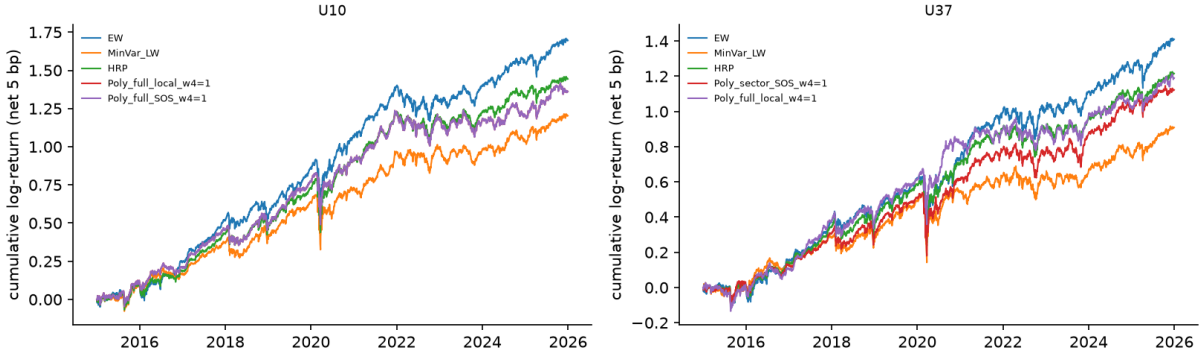


Figure 4: Experiment C. Cumulative log-return net of 5 bp for selected strategies, U10 (left) and U37 (right).

and local minimisers to coincide most of the time, and neither to be distinguishable from simple baselines [3].

*Observed.* On U10 the certified minimiser equals the local one on every date. On U37 the local solver is suboptimal by more than  $10^{-4}$  in relative terms on 14%, 6% and 4% of the dates, for  $w_4 = 0, 1, 3$ . Every SOS solve is certified.

Out of sample the picture is flat (Table 5, Figure 4). All 13 net-return intervals contain zero. The Sharpe-ratio test gives  $p$  between 0.32 and 0.97 (Appendix C, uncorrected).

On risk, the picture depends on the baseline. With  $w_4 = 1$  the U10 model has lower volatility than equal weights, by  $-1.0$  points (interval  $-2.0$  to  $-0.1$ ). It has higher volatility and CVaR than minimum variance and minimum CVaR.

On U37 the sector model is not better than the local solution of the same model. Its volatility differs by  $+0.19$  points ( $+0.06$  to  $+0.41$ ), so it is slightly higher.

The price of sparsity is large in model terms. The full-model objective at the sector solution is  $0.42$  on average against  $0.22$  at the full-model local solution ( $w_4 = 1$ ). It is worse on 100% of the dates.

The SOS solve takes 25 s per rebalance on U10 and 55 s on U37, against 27 s for the local solve of the sector model. These are wall times with six concurrent workers and are indicative only.

*Reading.* The sparse global minimiser is better at the thing it optimises, and the gain does not transfer. The most plausible reason is estimation error in the moments, which is what the protocol anticipated. I did not separate it from model error.

Table 5: Experiment C, out of sample 2015–2025 (132 monthly rebalances). Annualised volatility, daily 5% CVaR, maximum drawdown (%), mean one-way turnover per rebalance, effective number of assets, annualised return net of 5 bp, Sharpe ratio net of 5 bp.  $w_4 = 1$  for the polynomial models.

| universe | strategy          | vol (%) | CVaR (%) | MDD | turn. | $N_{\text{eff}}$ | net (%) | Sharpe |
|----------|-------------------|---------|----------|-----|-------|------------------|---------|--------|
| U10      | EW                | 16.4    | 2.43     | 36  | 0.04  | 10.0             | 15.5    | 0.94   |
| U10      | MinVar_LW         | 14.7    | 2.11     | 36  | 0.05  | 5.5              | 11.0    | 0.75   |
| U10      | ERC_LW            | 15.7    | 2.31     | 36  | 0.04  | 9.6              | 14.3    | 0.91   |
| U10      | HRP               | 15.3    | 2.24     | 35  | 0.11  | 8.1              | 13.2    | 0.86   |
| U10      | MinCVaR95         | 14.8    | 2.13     | 35  | 0.10  | 5.1              | 10.9    | 0.74   |
| U10      | Poly_full_local   | 15.4    | 2.23     | 34  | 0.14  | 4.6              | 12.4    | 0.81   |
| U10      | Poly_full_SOS     | 15.4    | 2.23     | 34  | 0.14  | 4.6              | 12.4    | 0.81   |
| U37      | EW                | 16.7    | 2.47     | 40  | 0.04  | 37.0             | 12.8    | 0.77   |
| U37      | MinVar_LW         | 14.4    | 2.06     | 40  | 0.12  | 9.3              | 8.3     | 0.58   |
| U37      | ERC_LW            | 15.7    | 2.32     | 38  | 0.04  | 34.4             | 11.9    | 0.76   |
| U37      | HRP               | 15.0    | 2.21     | 35  | 0.15  | 29.0             | 11.1    | 0.74   |
| U37      | MinCVaR95         | 14.6    | 2.05     | 40  | 0.23  | 7.2              | 8.2     | 0.56   |
| U37      | Poly_full_local   | 14.9    | 2.13     | 32  | 0.27  | 5.9              | 10.8    | 0.73   |
| U37      | Poly_sector_local | 14.7    | 2.14     | 36  | 0.07  | 18.6             | 10.1    | 0.69   |
| U37      | Poly_sector_SOS   | 14.9    | 2.17     | 36  | 0.11  | 17.9             | 10.2    | 0.69   |

## 6 Discussion and limits

What the experiments show. On synthetic polynomials, local search fails often enough that a certificate has value, especially above ten assets. The order-2 relaxation produces it in most cases, with a cost that exceeds 100 local runs at  $n = 50$ . On real data the certificate changes the allocation on few dates and does not improve on the risk-based baselines.

What they do not show. They do not show that the certificate is useless for other objectives. They do not show that the ball formulation fails for other uncertainty models, or with real estimation error instead of drawn error. They do not show anything about markets beyond 37 fixed US stocks over one period. The extrapolated dense costs at  $n = 30$  and  $n = 50$  are not measurements.

The bound-accuracy test covers validity and tightness at the  $10^{-8}$  level on small non-convex instances with two or three assets. It does not cover four assets or more, sparse relaxations with several cliques and auxiliary budget variables, or the real-data polynomials. For those I rely on the re-solve sample of Appendix B, which has no independent optimum, so a bias common to all solvers would go undetected.

Explanations I tested: the budget decoupling (fixed and confirmed by the regression test), and the solver accuracy behind the bound excess (Appendix B). Explanations I did not test: why local minima multiply with  $n$ , why shrinkage beats the ball in Experiment B, and why the global minimiser does not transfer out of sample. The synthetic polynomial has no return term and a simplified kurtosis structure, so Experiments A and B describe this family only.

*Note on tools.* Parts of the code and the text were written with an AI assistant. I reviewed the results, ran an audit of the claims, and corrected the errors it found (Appendix A and docs/CLAIMS.md).

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## A Corrections to version 1

Table 6 lists the claims of version 1 that did not survive, with the measured values. The register `docs/CLAIMS.md` gives the script and result file behind each line.

## B Accuracy of the bound

In Experiment C the bound exceeds the value of the same polynomial at the retained point, by a relative amount up to  $1.2 \times 10^{-4}$ . The sign is always the same, which first looked like an invalid relaxation. I re-solved 39 windows with CLARABEL tolerance  $10^{-9}$  instead of  $10^{-7}$ . Where the solver converges (no weight on the fourth moment, and the 2020 windows of U37) the excess falls by a factor 30 to 100. Where it stalls with an inaccurate status (most cases with  $w_4 \geq 1$ ) the median ratio of the two excesses is 1.00, and the excess stays between  $10^{-7}$  and  $1.5 \times 10^{-6}$  in most cases. The largest excess at tolerance  $10^{-9}$  is  $3.5 \times 10^{-6}$  and none is negative. The polynomial is the same on both sides (checked to nine digits) and the retained points are feasible (largest

Table 6: Claims of version 1 and their status.

| Claim in version 1                                                   | Measured                                                                                                                                         | Status         |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2-asset benchmark is non-convex                                      | Smallest Hessian eigenvalue on $K$ : +1.68; one local minimum                                                                                    | Refuted        |
| Default instances show the value of global search                    | Sparse bound, dense bound and local value coincide on the five default instances; one local minimum each                                         | Refuted        |
| Sparse bound equals dense bound                                      | Instance (6, 2, 0), skew 2: -22.1 against -2.43; instance (6, 2, 1), skew 4: -174 against -5.4. Cause: budget imposed on $E[\sum x]$             | Refuted, fixed |
| 398-fold reduction at $n = 50$                                       | 399 counts moment blocks only and leaves out the budget clique and the localising matrices; free moments give 253; metrics differed between rows | Corrected      |
| Dense solve $> 60$ s ( $n = 15$ ) and $> 60$ h ( $n = 50$ )          | Not SDP timings: they came from the scalar-loop builder                                                                                          | Unsupported    |
| Extraction for rank above one                                        | Random combination with non-orthogonal eigenvectors, atoms merged by position                                                                    | Rewritten      |
| Local solver returns a feasible bound                                | Scaling projection, success flag always true                                                                                                     | Rewritten      |
| “Distributionally robust (Delage–Ye)”                                | Frobenius ball on the kurtosis coefficients                                                                                                      | Renamed        |
| Guarantee “inaccessible to gradient methods”; real-time re-balancing | Promotional or unsupported                                                                                                                       | Removed        |

Table 7: Re-solves of a fixed sample (seed 2026) with CLARABEL at  $10^{-10}$  and SCS at  $10^{-8}$ , against the stored CLARABEL  $10^{-7}$  results. Largest relative change of the bound, relative to  $\max(1, |\text{lb}|)$  in B and C and to  $\max(10^{-3}, |\text{lb}|)$  in A. Flips: certification verdicts that change.

| exp. | cases | CLARABEL $10^{-10}$  | SCS $10^{-8}$        | verdict flips | other                                          |
|------|-------|----------------------|----------------------|---------------|------------------------------------------------|
| A    | 30    | $2.6 \times 10^{-5}$ | $8.8 \times 10^{-4}$ | 0 of 48       | failure flags changed: 0                       |
| B    | 30    | $4.5 \times 10^{-7}$ | $4.7 \times 10^{-7}$ | 0 of 60       | certification decision only                    |
| C    | 29    | $1.2 \times 10^{-4}$ | $1.3 \times 10^{-5}$ | 0 of 53       | largest allocation change $7.4 \times 10^{-7}$ |

violation  $4 \times 10^{-16}$ ). The reported value is the primal objective of the moment problem; the guaranteed bound would be the dual objective, which I did not extract. On a two-asset instance with a known optimum the excess is  $3.2 \times 10^{-9}$  and the solver residuals are about  $10^{-9}$ , at tolerance  $10^{-7}$  and  $10^{-10}$  alike. I conclude that the excess is a precision floor of the solver, consistent with an unfinished solve, and not a flaw in the relaxation.

**Sensitivity of the results.** I re-solved a fixed sample of 89 cases (seed 2026) with CLARABEL at  $10^{-10}$  and SCS at  $10^{-8}$  (Table 7). The sample includes inaccurate statuses, uncertified instances and the tightest certificate gaps of each experiment. Across the 161 re-solves, 0 certification verdicts change and 0 failure flags of Experiment A change. The SOS allocations of Experiment C move by at most  $7.4 \times 10^{-7}$ .

With CLARABEL, the bound itself changes by up to  $2.6 \times 10^{-5}$  (A),  $4.5 \times 10^{-7}$  (B) and  $1.2 \times 10^{-4}$  (C, the 2020 windows of U37). SCS differs by up to  $8.8 \times 10^{-4}$  on the 4-asset instances of A, where it also reports an inaccurate status. Its flatness decisions differ from CLARABEL’s in 8 of 24 cases of C, so I do not use it for rank decisions. The 5 instances where the sparse bound is more than  $10^{-3}$  below the dense one stay so with CLARABEL at  $10^{-10}$ . The difference lies between  $1.2 \times 10^{-3}$  and  $9.0 \times 10^{-3}$  and changes by at most  $1.1 \times 10^{-5}$ , so the weaker sparse bound is not a solver artefact.

I treat a change as harmless when it cannot alter a verdict or a figure of the paper. Verdicts use a relative tolerance of  $10^{-4}$ . The quantities that carry conclusions (verdicts, failure flags, allocations, dense-minus-sparse differences) change by at most  $1.1 \times 10^{-5}$  or not at all. The bound values change by more than that on some cases, up to  $1.2 \times 10^{-4}$ , without altering any verdict. I therefore did not rerun Experiments A, B and C.

**Independent optimum.** A test (`tests/test_bound_soundness.py`) compares the bound with a minimum computed without the SDP on 6 instances with two or three assets, by a fine grid followed by local refinement. It also checks that the bound stays below  $f$  at 2000 random feasible points, at the local solution and at the extracted points. At tolerance  $10^{-10}$  the difference between the bound and the independent minimum lies between  $-8.0 \times 10^{-9}$  and  $3.2 \times 10^{-9}$ . A tolerance of  $10^{-9}$  cannot be enforced because the solver stalls with residuals of that order. The test therefore allows  $+10^{-8}$  (bound above the optimum) and  $-5 \times 10^{-8}$  (bound far below it).

## C Experiment C: all comparisons

Table 8: Experiment C, all 13 pairwise comparisons, difference A minus B in percentage points (annualised volatility; daily 5% CVaR; annualised net return at 5 bp) with 95% circular-block bootstrap intervals (blocks of 21 days), and the p-value of the Ledoit–Wolf Sharpe-ratio test. No multiple-comparison correction; the intervals are descriptive.

| U   | A          | B            | vol                  | CVaR                 | net return        | $p$  |
|-----|------------|--------------|----------------------|----------------------|-------------------|------|
| U10 | SOS full   | EW           | -1.00 [-1.96, -0.14] | -0.20 [-0.32, -0.08] | -3.1 [-7.5, +1.4] | 0.33 |
| U10 | SOS full   | MinVar       | +0.72 [+0.25, +1.20] | +0.12 [+0.02, +0.21] | +1.4 [-1.4, +4.2] | 0.60 |
| U10 | SOS full   | ERC          | -0.28 [-1.10, +0.47] | -0.08 [-0.19, +0.03] | -1.9 [-5.5, +1.8] | 0.38 |
| U10 | SOS full   | HRP          | +0.06 [-0.64, +0.72] | -0.02 [-0.11, +0.08] | -0.8 [-3.8, +2.3] | 0.62 |
| U10 | SOS full   | MinCVaR      | +0.58 [+0.27, +0.92] | +0.10 [+0.03, +0.17] | +1.5 [-1.1, +4.1] | 0.49 |
| U10 | SOS full   | local full   | +0.00 [-0.00, +0.00] | +0.00 [-0.00, +0.00] | +0.0 [-0.0, +0.0] | 0.71 |
| U37 | SOS sector | EW           | -1.80 [-2.22, -1.36] | -0.29 [-0.39, -0.19] | -2.6 [-5.9, +0.7] | 0.43 |
| U37 | SOS sector | MinVar       | +0.48 [-0.06, +1.05] | +0.11 [+0.01, +0.20] | +1.9 [-1.0, +5.2] | 0.32 |
| U37 | SOS sector | ERC          | -0.84 [-1.15, -0.45] | -0.15 [-0.22, -0.06] | -1.7 [-4.0, +0.6] | 0.38 |
| U37 | SOS sector | HRP          | -0.18 [-0.40, +0.09] | -0.03 [-0.09, +0.03] | -0.9 [-2.6, +1.2] | 0.49 |
| U37 | SOS sector | MinCVaR      | +0.31 [-0.33, +0.97] | +0.12 [+0.01, +0.22] | +2.0 [-1.4, +5.6] | 0.33 |
| U37 | SOS sector | local sector | +0.19 [+0.06, +0.41] | +0.03 [-0.00, +0.08] | +0.1 [-1.2, +1.5] | 0.97 |
| U37 | SOS sector | local full   | -0.01 [-0.52, +0.45] | +0.05 [-0.05, +0.16] | -0.6 [-4.2, +2.9] | 0.76 |

## D Reproduction

Software: Python 3.12.3, CVXPY 1.9.3, CLARABEL 0.11.1, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.6. Machine: Intel Core i5-10310U CPU @ 1.70GHz, 8 threads; experiments used six single-threaded workers. The package installs with `pip install -e .`; tests run with `pytest -m "not slow"`. Experiment protocols are in `experiments/PROTOCOL*.md`, written before the runs. Protocol C has three amendments, all made before the evaluation period: a weight cap of 0.25 in place of 1 for conditioning, an asset-level normalisation of the moment weights, and a cap of 64 on atom combinations in the extraction (disjoint sector cliques otherwise gave about  $2 \times 10^6$  candidates). Experiments: `experiments/exp_a_local_vs_global.py`, `exp_b_robust.py`, `exp_c_real_data.py`, each followed by its `summarize_*.py`. The prices come from `scripts/download_data.py`. All numbers and tables of this paper are generated by `scripts/make_paper_numbers.py` from `results/`. Experiment C ran for about two hours on six workers.